

Cognitive Task Overview: TestMyBrain Trail-Making

Contact: Info@ManyBrains.net

ManyBrains.net

TestMyBrain.org

TMB Test Name: TestMyBrain Trail-Making

Test Demo: [Trail-Making A](#) | [Trail-Making B](#)

Document Version: May.22.2026

The Many Brains Project

[The Many Brains Project](#) is a US-based 501(c)3 non-profit focused on the development of digital cognitive testing tools. We currently support many different types of research studies through our infrastructure for cognitive assessment - these range in size from small lab-based pilot studies to large longitudinal, multisite clinical research studies with tens of thousands of participants. As [TestMyBrain.org](#) has been continuously in operation since 2008, we provide a stable and secure platform for hosting and delivering mobile and web-based cognitive assessment protocols. Through TestMyBrain.org, data have been collected from over 2.5 million participants in a *citizen science* framework that includes structured return of research results toward the development, validation, and normative characterization of cognitive measures. We currently support research and education at over 400 sites worldwide as well as support for over 1200 clinicians or clinical sites engaged in remote digital neuropsychological assessment. For more information contact info@manybrains.net.

CITATION

Please credit The Many Brains Project and TestMyBrain in any papers, posters, or publications related to the TMB tests or data collected by TMB tests.

- Example:
 - All tasks were selected from and hosted on The Many Brains Project's web-based cognitive testing platform, TestMyBrain (Germine et al., 2012; The Many Brains Project).
 - Germine, L., Nakayama, K., Duchaine, B. C., Chabris, C. F., Chatterjee, G., & Wilmer, J. B. (2012). Is the Web as good as the lab? Comparable performance from Web and lab in cognitive/perceptual experiments. *Psychonomic Bulletin & Review*, 19(5), 847-857.
 - The Many Brains Project. *TestMyBrain Cognitive Tests*. URL: www.manybrains.net

Test Overview

Background:

In TMB Trail-Making (Singh et al., 2021; Treviño et al., 2021), modeled after traditional pencil and paper trail-making assessments (Bowie & Harvey, 2006), participants are asked to sequentially connect target dots as quickly and accurately as possible. Trail-Making A measures processing speed, while Trail-Making B additionally measures task switching and cognitive flexibility.

Task Parameters:

In TMB Trail-Making A, participants connect targets labeled by consecutive numeric digits (e.g., 1-2-3-4). In TMB Trail-Making B, participants alternate between number and letter labels to connect target dots (e.g., 1-A-2-B). When incorrectly connecting target dots, participants are required to go back to the most recently correctly selected target, and then draw a path to the next correct target before continuing with the test. In both versions A & B, participants complete a portion of a trail as practice, then complete one scored test trail.

Primary Outcome:

The suggested primary outcome for both test versions is the median reaction time between connecting each consecutive target. Researchers may also consider incorporating the total time taken to complete each trial, as well as the number of errors made by each participant.

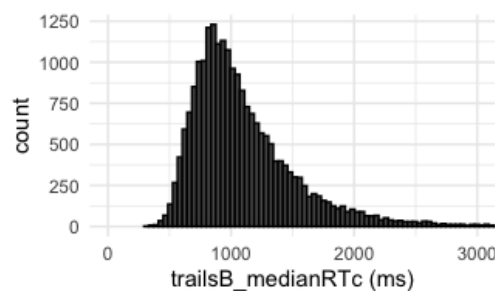
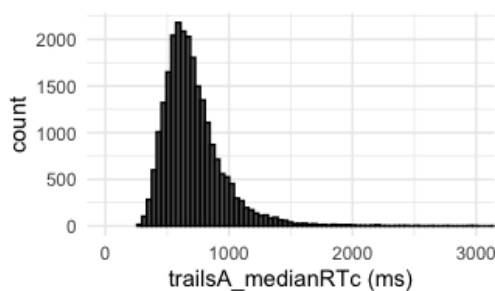
User Input:

Participants must use either a mouse or touchscreen input to connect the dots. Note that this test is more challenging to complete on small devices (Passell et al., 2021), so completing the test using a smartphone is not recommended.

Alternate Task Versions: Alternate forms of the test are available for repeated administration.

Psychometrics:

- **Reliability:** In single-session testing, between-participant variation in performance (median correct reaction time) has a split-half reliability (first half versus second half of test) of .84 for Trail-Making A and .68 for Trail-Making B.
- **Score distribution:**



References:

- Passell, E., Strong, R. W., Rutter, L. A., Kim, H., Scheuer, L., Martini, P., ... & Germine, L. (2021). Cognitive test scores vary with choice of personal digital device. *Behavior Research Methods*, 1-14.
- Singh, S., Strong, R. W., Jung, L., Li, F. H., Grinspoon, L., Scheuer, L. S., Passell, E. J., Martini, P., Chaytor, N., Soble, J. R., & Germine, L. (2021). The TestMyBrain Digital Neuropsychology Toolkit: Development and Psychometric Characteristics. *Journal of Clinical and Experimental Neuropsychology*, 43(8), 786-795.
- Treviño, M., Zhu, X., Lu, Y. Y., Scheuer, L. S., Passell, E., Huang, G. C., ... & Horowitz, T. S. (2021). How do we measure attention? Using factor analysis to establish construct validity of neuropsychological tests. *Cognitive Research: Principles and Implications*, 6(1), 1-26. <https://doi.org/10.1186/s41235-021-00313-1>